

TrES-3b

R-1524

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190410 · created 2026-08-01T17:29:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458583.8403 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.02
Transit depth (Rp/Rs)^2	3.82 +/- 0.77 [%]
Orbital Inclination (inc)	81.36 +/- 0.43
Airmass coefficient 1 (a1)	1.038 +/- 0.012
Airmass coefficient 2 (a2)	-0.0211 +/- 0.004
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	#1 - [300, 135]
Optimal Aperture	3.31
Optimal Annulus	13.58
Transit Duration (day)	0.054 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.02 vs 0.16309 (deviation 1.25σ)
Duration fit vs published	0.054 d vs 0.0567 d (deviation 0.42σ)
Mid-time O–C	-0.9 min
Depth SNR	7.7
Residual scatter	1.2 %

Light curve

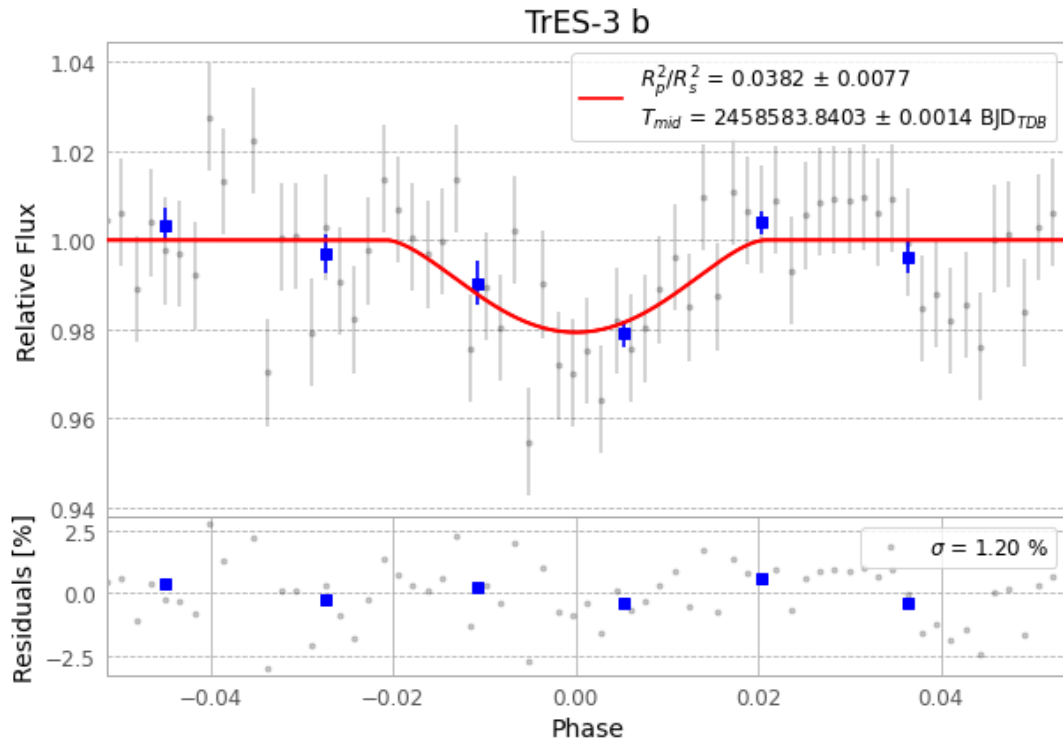


Figure 1 — FinalLightCurve_TrES-3 b_2019-04-09.png (SHA-256 84762d8cc15fe541...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 50], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a4a943c2167e40af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), TRES-3190410062743.FITS → TRES-3190410094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190410.log (3f12aa889be6...), FinalLightCurve_TrES-3 b_2019-04-09.png (84762d8cc15f...), FinalLightCurve_TrES-3 b_2019-04-09.csv (4ef9fd3f72c2...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:29Z.